Mrinaal Ramachandran

M.S. Computer Science, University of Massachusetts Amherst
Google Scholar · LinkedIn · GitHub · Website · ORCID

PROFILE

My main professional focus is computational criminology: using formal modeling and computational methods to understand technology-facilitated exploitation, and translating research into intelligence and tools for prevention and disruption. I'm currently a Strategic Research Fellow at Project VIC International, where I'm conducting research into the mechanics of technology-mediated exploitation and contributing to the CASE/UCO and Crimes Against Children Ontology open-source infrastructure.

RESEARCH EXPERIENCE

CaseLinker: Open-Source Research Platform

Oct 2025 – Present

Independent Researcher, building a tool for knowledge representation and cross-case analysis of Internet Crimes Against Children (ICAC) cases

- Built a corpus of **10,362 case reports** spanning **125,891 extracted features** and **4,137 law enforcement agencies** across 56 public ICAC sources (ICAC Task Forces, DOJ, NCMEC, and state AG offices), compiled entirely from public records under UMass HRPO NHSR Determination #7668
- Implemented deterministic feature extraction so every feature traces back to source text, making cases comparable across technology platforms, agencies, jurisdictions, and prosecutorial outcomes
- Mapped the corpus onto a semantic knowledge graph aligned to the CAC Ontology and CASE/UCO standard, with SPARQL and natural-language search across cases; built an MCP server exposing **52 tools** for agent search, analysis, and collection (server)
- Developed the *Affordances for Harm* framework, identifying recurring platform capabilities across technology generations; proposed a Foreseeable Harm Standard for assessing new platform features before deployment
- Corpus adopted by Project VIC International into its production knowledge graphs
- **Code and Technical Reports:** GitHub · CaseLinker: An Open-Source System for Cross-Case Analysis of ICAC Reports · Interpretable ML Approaches for Analyzing ICAC Reports · Painting the Landscape of Internet Crimes Against Children with Interpretive Tooling

Project VIC International

Sept 2026 – Present

Strategic Research Fellow (Research Collaborator, Jun – Aug 2026)

- Developing the Exploitation State Machine, an MDP-based formal model of exploitation trajectories; research protocol cleared under UMass HRPO NHSR Determination #8252
- Contributed extensions to the CAC Ontology, Project VIC's Crimes Against Children domain ontology, including state-machine classes for ICAC exploitation-lifecycle modeling: the AffordanceMisuse class (PR #33, merged) and the ConditioningPhase macro-phase with a 9-value controlled vocabulary (PR #39, merged)
- Authored the trajectories extension to the CASE-UCO-SDK, enabling exploitation-pathway modeling within the CASE/UCO cyber-investigation standard, and a public-record capture recipe; both shipped in SDK v1.31.0 after review across #86, #87, #104, #135, and #145
- Conducted Project VIC-supported collection and analysis of **30 primary-source federal court records** (PACER), including indictments, Statements of Offense, and sentencing memoranda, to construct the exploitation state machines underlying the *Affordances for Harm* publication
- Direct two research interns on CaseLinker collection and data-analysis work, including corpus review, deduplication, and code review on public repositories
- Nominated by Cory Hall to the Cyber Domain Ontology Technical Steering Committee for ontology contributions

PERSUE Labs, Arizona State University

Feb 2024 – May 2025

Undergraduate Research Assistant, Tempe, AZ

- Investigated data access and security implications of Zoom mini-apps under Professor Hasan
- Co-authored *Who's Watching You Zoom? Investigating Privacy of Third-Party Zoom Apps*, published at PETS 2025, from a year-long longitudinal study analyzing **97,000+ snapshots** of Zoom Marketplace apps
- Built a research testbed app with the Zoom SDK to study data access and manipulation capabilities firsthand
- Performed privacy and security analysis of Zoom Marketplace apps via API reverse-engineering, OAuth scope auditing, and Wireshark analysis

PUBLICATIONS

Peer-Reviewed

- Goenka, S., Prabhu, A., Sakurai, P., Ramachandran, M., & Hasan, R. (2025). *Who's Watching You Zoom? Investigating Privacy of Third-Party Zoom Apps*. Proceedings on Privacy Enhancing Technologies, 2025(4), 530–548. doi.org/10.56553/popets-2025-0143

Preprints

- Ramachandran, M. (2026). *Affordances for Harm: How Offenders Misuse Platform Capabilities to Exploit Children, and Where to Intervene*. Zenodo. doi.org/10.5281/zenodo.21347781
- Ramachandran, M. (2026). *CaseLinker: An Open-Source System for Cross-Case Analysis of Internet Crimes Against Children Reports*. arXiv:2603.18020; Zenodo: doi.org/10.5281/zenodo.18744216

COURSE RESEARCH PROJECT

Agent Tool Calling and Evidence-Graph Construction

Fall 2026 – Present

COMPSCI 684, Trustworthy and Responsible AI (Prof. Eugene Bagdasarian), four-person group project

- Improving how LLM agents call and chain tools, process evidence, and produce CASE/UCO evidence graphs through the CASE-UCO-SDK's MCP server, on local and frontier models
- Ran a baseline and filed failures upstream: entity over-typing in document processing (#158), routing abstention that drops ranked recipe hits (#159), and critic finalization that accepts empty manual reviews (#161); fixed #158 with a NANP plausibility gate on phone-number typing (PR #160)

TECHNICAL EXPERIENCE

DriverAI (Startup)

Jun – Aug 2025

Android Security Intern, Remote

- Developed secure authentication workflows using SQLCipher, Play Integrity API, and Credential Manager for an Android MVP app, with an AES encryption layer over the Room database for sensitive user input
- Implemented JWT-based token verification integrated with Firebase Auth and AWS Cognito; conducted root detection and SELinux status checks across device profiles, including emulator bypass techniques
- Collaborated directly with the startup's CTO to define milestones and scope future security work

Ultra-Wideband (UWB) Data Communication Capstone

General Dynamics Mission Systems

CSE 485/486, Computer Science Capstone Project, Arizona State University

- Designed and implemented UWB communication boards for bidirectional data transmission with modular, scalable code
- Validated UWB feasibility through hardware testing (signal strength, latency, reliability) with GDMS engineers across a two-semester Agile cycle

SKILLS

- **Engineering:** Python, Java, scikit-learn (random forests), LLMs, agents, and tool calling, Model Context Protocol, local inference (Ollama), agentic coding (Claude Code, Cursor), system architecture and design
- **Knowledge Representation:** CASE/UCO and CAC ontologies, OWL/RDF, SHACL validation, SPARQL, deterministic information extraction, cross-case similarity (weighted Jaccard), affordance modeling
- **Research Methods:** Markov decision process modeling, federal court-record analysis (PACER/RECAP), public-records corpus construction, NHSR/IRB protocols, API reverse-engineering, OAuth scope auditing, Wireshark

EDUCATION

University of Massachusetts Amherst

Expected May 2027

M.S. in Computer Science

- **Relevant Coursework:** Trustworthy and Responsible AI, Machine Learning, Systems for Data Science, Digital Forensics, Advanced Computer Network Security, Internet Law, Mobile and Wireless Networks

Arizona State University

2022 – 2025

B.S. in Computer Science

- **Relevant Coursework:** Operating Systems, Computer Architecture, Information Assurance, Statistics, Linear Algebra, Data Structures and Algorithms, Distributed Software Development, Artificial Intelligence, Natural Language Processing

LEADERSHIP & ACHIEVEMENTS

FIRST Tech Challenge, Captain, Tacobots

2017 – 2022

- Led the team through two FIRST World Championship qualifications (St. Louis and Houston), winning awards at both
- Developed FTC SDK software on the Android platform in Java; applied engineering design process and led team outreach

FIRST, Volunteer

2017 – 2019

- Organized regional competitions and recruited volunteers, contributing 100+ hours per year; served as Master of Ceremonies

Eagle Scout, Boy Scouts of America (2021)